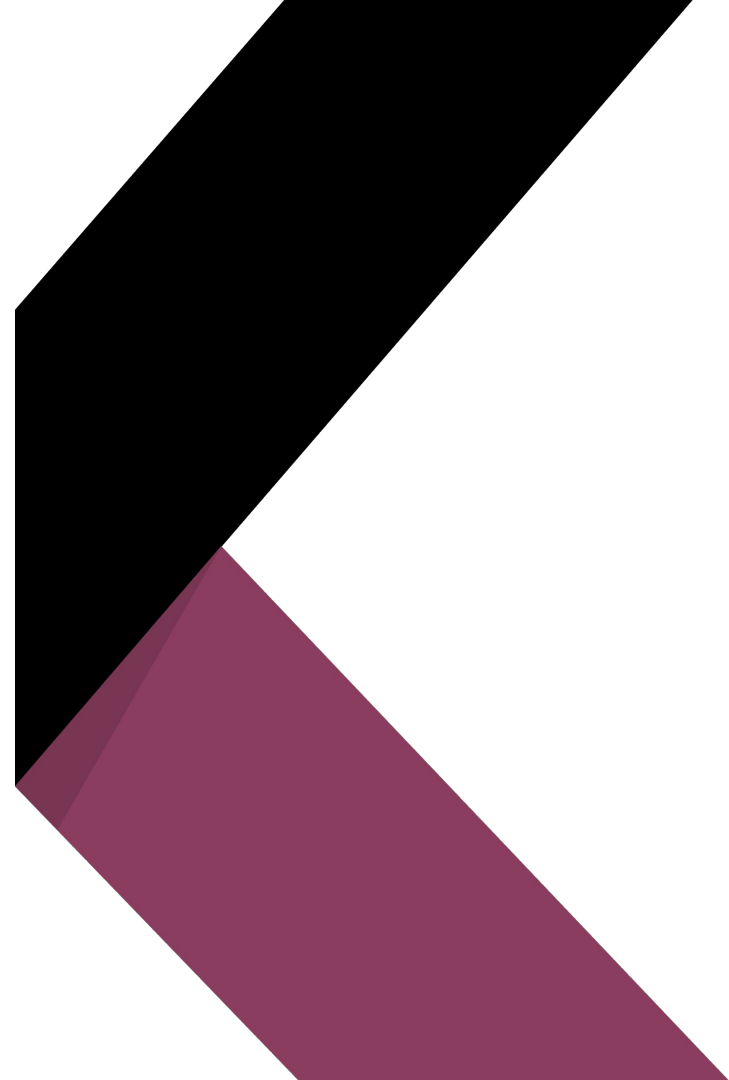


Diagnosis

DTS Monaco



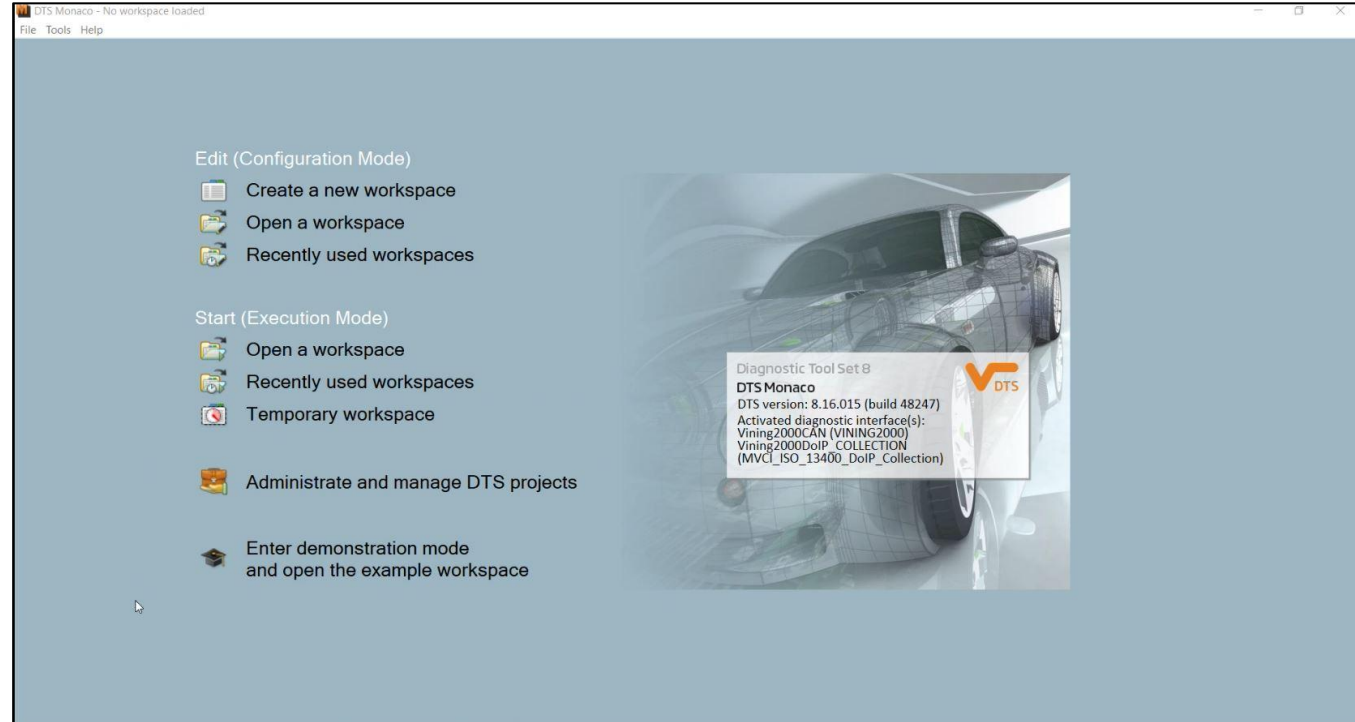


DTS Monaco



DTS Monaco - Introduction

Softing DTS.monaco is an extensive offboard diagnostic tool for the engineering sector which covers the entire range of application cases from ECU testing through to vehicle release. It is easily integrated into the test sequences and corporate processes, not least because of its flexible interfaces which can be configured to suit the relevant test step





Areas of Application

- Development of diagnostic and control functions for ECUs
- Function test and validation
- Integration and system test
- Preparation of test sequences for Manufacturing and Service
- Analysis of returns and Quality Assurance

Advantages

- Reduced costs and familiarization time as DTS Monaco covers the functionality of several tools, previously separate: OBD scan tool, data and bus monitor
- Fast results thanks to intuitive operation and preconfigured templates
- Top quality thanks to early detection and remedy of communication problems and function errors
- Highly effective thanks to flexible adaptability to various tasks
- Communication data and test results can be fully documented



DTS Monaco - Hardware Configuration

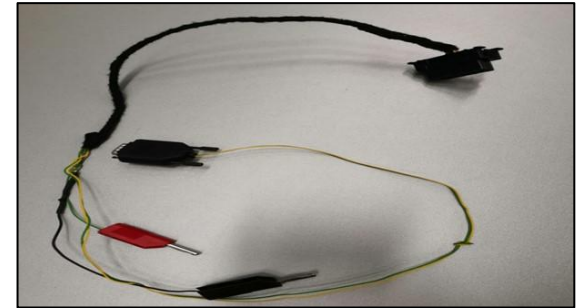
VIN|ING 2000



Dev Connector



Split Connector





DTS Monaco - VIN|ING 2000

VIN|ING 2000 is a further powerful VCI for the VIN|ING product family. With a compact design and WLAN, LAN and USB as interfaces to the host system as well as CAN/FD, K-Line and Ethernet to the vehicle, VIN|ING 2000 is particularly well suited for future-proof manufacturing and after-sales service applications.

The WiFi interface of the VIN|ING 2000 is equipped with two separate communication channels and with IEEE 802.11 a/b/g/h/n supports the 2.4 and 5 GHz band. Current encryption technologies such as WPA2/PSK and WPA2/RADIUS as well as high-performance roaming characteristics are the prerequisite for use on the production line or in after-sales service. Furthermore the device has various Sleep/WakeUp modes and programmable function keys for interaction in diagnostic sequences. When communicating with the host PC over USB or LAN, the Mag- Code connection is a predetermined breaking point, which separates the cable connection in the case of a considerable mechanical load



DTS Monaco - System Configurator

System Configurator

File Edit View System Project Help

System settings Project administration Libraries

Directories
Language
Interfaces
Java
Licenses
Trace
Runtime system properties

Simulation (Simulation) Connected SN: Not Available

Vining2000CAN (Vining2000)

Interface name	Vining2000CAN
Bus system interface	D-PDU API
D-PDU API version	Softing AG EDIC_D_PDU_API_11_30_018
Module type	Vining2000
Vendor module name	VendorName='Softing AG' ModuleName='Vining2000' SerialNumber='005002219'
Cable	---
Use for licensing	VendorName='Softing AG' ModuleName='Vining2000' SerialNumber='005002219'
Physical links	VendorName='Softing AG' ModuleName='Vining2000' SerialNumber='005002533'
ISO 11898_2_DWCAN (6 HL 14 LOW)	VendorName='Softing AG' ModuleName='Vining2000' SerialNumber='005002548'
ISO 11898_2_DWCAN (12 HL 13 LOW)	CAN 2

Vining2000DoIP_COLLECTION (MVCI_ISO_13400_DoIP_Collection)

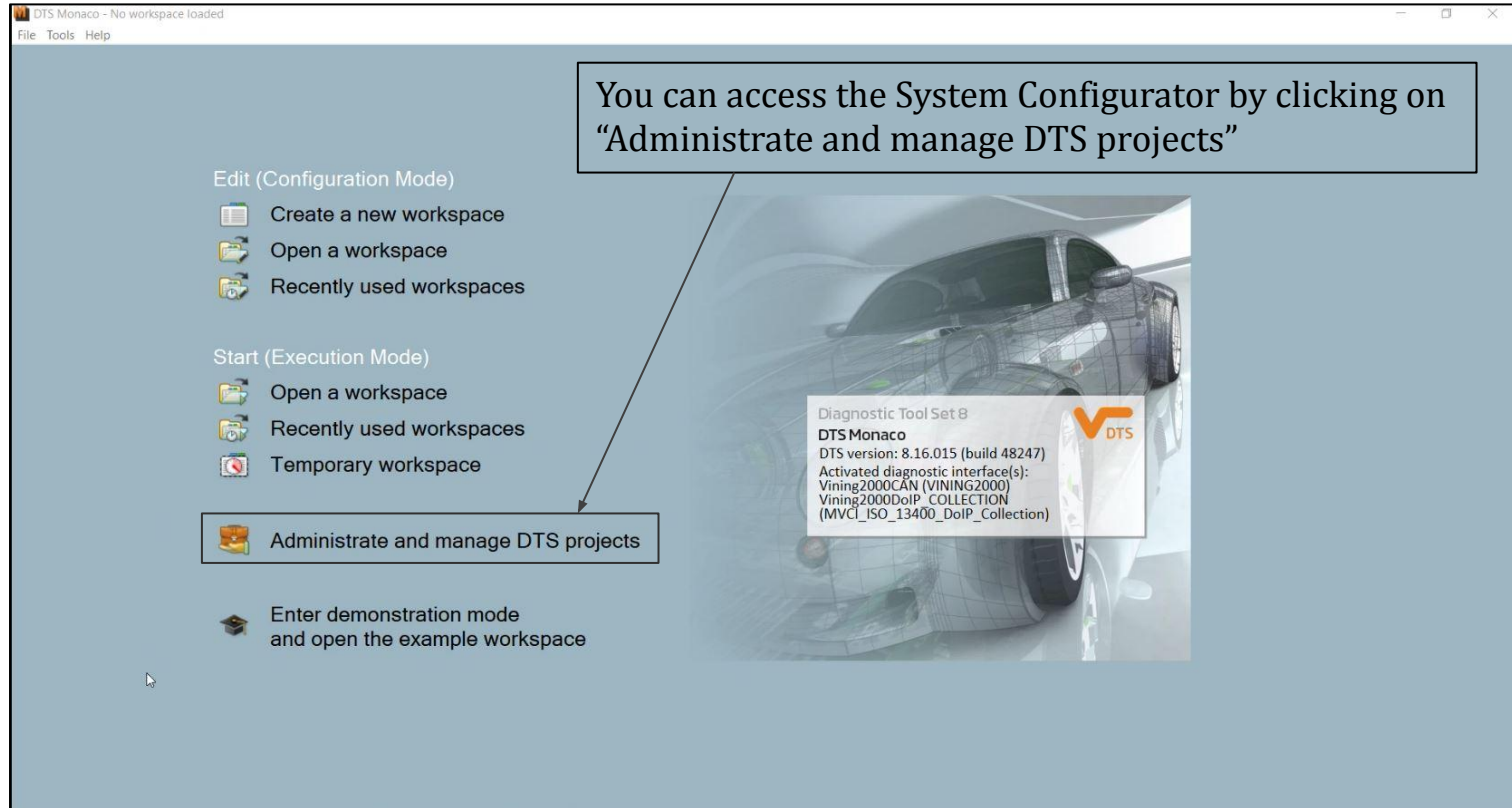
EDICUS8 (EDICust8)

Ready CAP NUM SCR

First step is to connect your laptop to the WiFi network of the VIN|ING 2000 and then with the System Configurator you have to chose the VIN|ING 2000 serial number that you are using to link it to DTS Monaco on your laptop

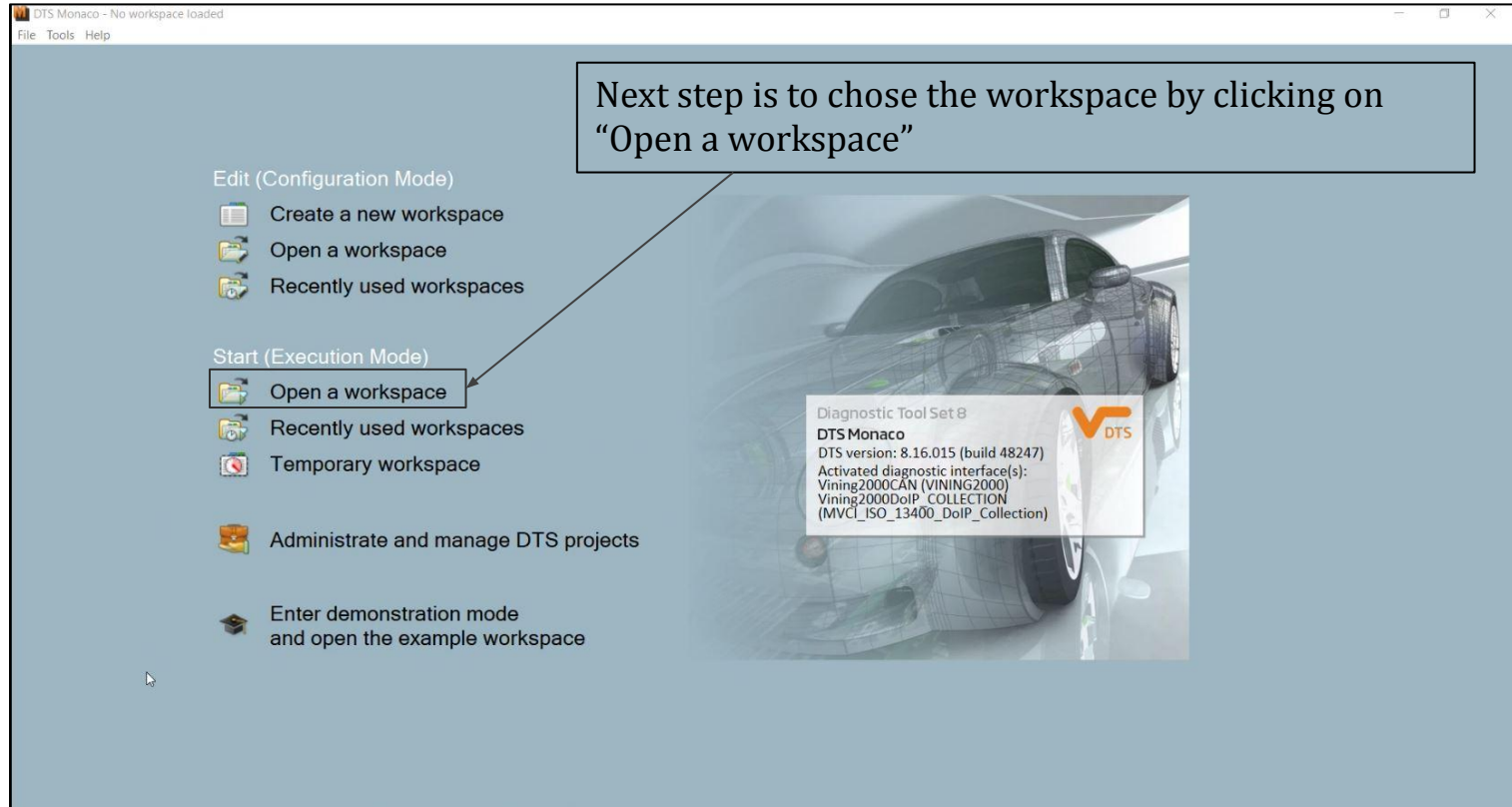


DTS Monaco - System Configurator





DTS Monaco - Workspace selection





DTS Monaco - Workspace selection

Here you can chose the Project and the specific Workspace to use

The screenshot displays the DTS Monaco application window with the title bar "DTS Monaco - No workspace loaded". The menu bar includes "File", "Tools", and "Help". The main interface is divided into several sections. On the left, there are icons and labels for "Edit (Configuration)", "Start (Execution Manager)", and "Enter demonstration mode and open the workspace". The "Edit (Configuration)" section includes "Create a new workspace", "Open a workspace", and "Recently used workspaces". The "Start (Execution Manager)" section includes "Open a workspace", "Recently used workspaces", "Temporary workspace", "Administrative workspace", and "Enter demonstration mode and open the workspace". The "Open Workspace" dialog box is open, showing a list of projects and workspaces. The "Projects" pane lists "P16_v2_Dev" and "P16_v2_Prod". The "Workspaces" pane lists various workspaces, with "4) Clear DTC and DTC Read for all and Each unit on DOIP ECUs - J250p" selected. The dialog box has "OK", "Cancel", and "Help" buttons at the bottom.



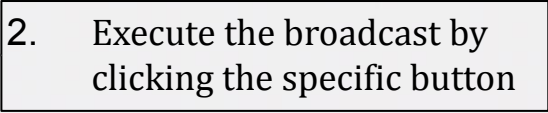
DTS Monaco - DoIP Connection

1. Execute the Ethernet activation by clicking the specific button

The screenshot shows the DTS Monaco software interface for production general diagnostics. The top toolbar contains various icons, with one circled and pointed to by a callout box. The interface includes a 'Bus Trace' section on the left, a 'Diagnostic Services' section in the center, and a 'Process Values' table at the bottom. The 'Diagnostic Services' section shows a search for 'ACF' and lists 'ACF (CAN_ACF)' and 'ACF (DoIP_ACF)'. The 'Process Values' table has columns for Name, Value, Unit, RangeInfo, and Error.

Ready

Active Execution Mode: DoIP ECU available (7) NUM







DTS Monaco - Diagnostic Workspace

DTS Monaco - 1) Production General Diagnostics_Doip [P16_v2_Prod]

File Edit View Execution Tools Help

14.10 V

Bus Trace

Now monitoring on CAN1

Rel. Tr... ID Data

ECUs tab menu

SCMD SCMP EDPC SAS ADI MCU ECM AMP IFU TCU IBS DLL DLR BLRF BLRR FWM ZA1 ZB1 ZB2 ZC1 ESW BMS Commu...

Shortcut buttons

ECU Reset

Clear DTCs

DTC Logging ON

DTC Logging OFF

Diagnostic Services

Search

Global Search

<Please enter>

Diagnostic

P16_ZA1_DCU_3_5 (DOIP_ZA1)

P16_ZA1_DCU_3_5 (DOIP_ZA1) ZA1_DCU_VARIANT_00

Diagnostic services window

Settings Description

Logical Link P16_ZA1_DCU_3_5 (DOIP_ZA1) ZA1_DCU_VARIANT_00

Property Value Unit

Diagnostic settings and description window

PDU:

Current selection

Transmit

Cyclic (ms) 2000

Transmit preconfigured

Extended Session

Level 1 Access

Program Session

Level 2 Access

Shortcut buttons

Process Values

Name	Value	Unit	RangeInfo	Error
[Tester]	[e00]			
StartCommunication	eALL_POSITIVE			

Diagnostic Trace window

[15:17:43.828] Type="MVC_ISD_DoIP_Collection" Master_MVCI="VendorName="Softing AG" ModuleName="VINING2000" SerialNumber="005002445" Count="7" LA0="0x07C4" IP0="1d93:1" P16_v2_Prod / DefaultVIT

Ready

Active Execution Mode DoIP ECU available (7) NUM



DTS Monaco - Diagnostic Services Window

DTS Monaco - 1) Production General Diagnostics_Doip [P16_v2_Prod]

File Edit View Execution Tools Help

14.10 V

Bus Trace
ORC
Now monitoring on CAN1

Rel.TL... ID Data

Diagnostic Services

Search ☐ Global Search

<Please enter>

Diagnostic

- [09] ECUNAME Read InfoType Data (OBD)
- [09] Supported InfoTypes Read supported InfoTypes (OBD)
- [0A] Emission-related Trouble Codes Read (Permanent Status)
- [10] Default Session Start
- [10] Default Session Start_NoResponse
- [10] Extended Session Start
- [10] Extended Session Start_NoResponse
- [10] Programming Session Start
- [10] Programming Session Start_NoResponse
- [11] Hard Reset
- [11] Soft Reset
- [14] Fault Memory Clear
- [19] Fault Memory Read By Status Mask
- [19] Fault Memory Read Extended
- [19] Fault Memory Read Number By Status Mask
- [19] Fault Memory Read Snapshot
- [19] Fault Memory Read Supported
- [22] AD Controller Inputs Read
- [22] AD Faults Read
- [22] AD Force Demands Read
- [22] AD Sensor Inputs Read
- [22] AD Valve Drives Read
- [22] AN1 Read

Settings Description

Logical Link P16_ZA1_DCU_3_5 (DOIP_ZA1) ZA1_DCU_VARIANT_00

Property	Value	Unit
----------	-------	------

PDU:

Current selection Transmit ☐ Cyclic (ms) 2000 Transmit preconfigured Extended Session Level 1 Access Program Session Level 2 Access

Process Values

Name	Value	Unit	RangeInfo	Error
------	-------	------	-----------	-------

[15:17:43.828] Type="MVD_ISO_13400_DoIP_Collection" Master_MVCI="VendorName="Softing AG" ModuleName="VINING2000" SerialNumber="005002445" Count="7" LA0="0x07C4" IP0="1d93:1" P16_v2_Prod / DefaultVMT

Ready Active Execution Mode: DoIP ECU available (7) NUM



DTS Monaco - Diagnostic Settings and Description Window

DTS Monaco - 1) Production General Diagnostics_Doip [P16_v2_Prod]

File Edit View Execution Tools Help

14.08 V

Bus Trace ORC Configure...

Now monitoring on CAN!

Rel.Tl... ID Data

Diagnostic Services

Search Global Search

<Please enter>

Diagnostic

- [09] ECUNAME Read InfoType Data (OBD)
- [09] Supported InfoTypes Read supported InfoTypes (OBD)
- [04] Emission-related Trouble Codes Read (Permanent Status)
- [10] Default Session Start
- [10] Default Session Start_NoResponse
- [10] Extended Session Start
- [10] Extended Session Start_NoResponse
- [10] Programming Session Start
- [10] Programming Session Start_NoResponse
- [11] Hard Reset
- [11] Soft Reset
- [14] Fault Memory Clear
- [19] Fault Memory Read By Status Mask
- [19] Fault Memory Read Extended
- [19] Fault Memory Read Number By Status Mask
- [19] Fault Memory Read Snapshot
- [19] Fault Memory Read Supported
- [22] AD Controller Inputs Read
- [22] AD Faults Read
- [22] AD Force Demands Read
- [22] AD Sensor Inputs Read
- [22] AD Valve Drives Read
- [22] ANT Read

Settings Description

Logical Link P16_ZA1_DCU_3_5 (DOIP_ZA1) ZA1_DCU_VARIANT_00

Diagnostic Service [19] Fault Memory Read By Status Mask

Property	Value	Unit
Params		
SID-RQ	0x19	
ReportDtcByStatusMask	0x02	
DTC Status Byte	<STRUCTURE>	
Test failed	<invalid>	
Test failed this monitoring cycle	<invalid>	
Pending DTC	false	
Confirmed DTC	true	
Test not completed since last clear	false	
Test failed since last clear	true	
Test not completed this monitoring cycle	<invalid>	
Warning indicator requested	<invalid>	

PDU: (1/3) 19.02.08

Current selection Transmit Cyclic (ms) 2000 Transmit preconfigured Extended Session Level 1 Access Program Session Level 2 Access

Process Values

Name	Value	Unit	RangeInfo	Error
------	-------	------	-----------	-------

[15:17:43.828] Type=MVCL_ISO_13400_DoIP_Collection Master_MVCL=VendorName=Softing AG ModuleName=VINING2000 SerialNumber=005002445 Count=7 LA0=D07C# IP6=Id93: P16_v2_Prod / DefaultMT

Ready Active Execution Mode: DOIP ECU available (7) NUM



DTS Monaco - Diagnostic Trace Window

DTS Monaco - 1) Production General Diagnostics_Doip [P16_v2_Prod]

File Edit View Execution Tools Help

14.08 V

Bus Trace
Now monitoring on CAN1
Rel.Tl... ID Data

SCMD SCMP EDPC SAS ADI MCU ECM AMP IFU TCU IBS DLL DLR BLRF BLRR FWM ZA1 ZB1 ZB2 ZC1 ESW BMS Comm...

ECU Reset
Clear DTCs
DTC Logging ON
DTC Logging OFF

Diagnostic Services
Search
Global Search
«Please enter»

Diagnostic
P16_ZA1_DCU_3_5 (DOIP_ZA1) ZA1_DCU_VARIANT_00
P16_ZA1_DCU_3_5 (DOIP_ZA1) ZA1_DCU_VARIANT_00
Hex Services
Protocol Parameter Sets
ComPimilives
[01] Accelerator Pedal Position D Read PID (OBD)

Settings Description
P16_ZA1_DCU_3_5 (DOIP_ZA1) ZA1_DCU_VARIANT_00

Property Value Unit
PDU: (1/3)

Current selection
Transmit Cyclic (ms) 2000
Transmit preconfigured
Extended Session Level 1 Access Program Session Level 2 Access

Process Values

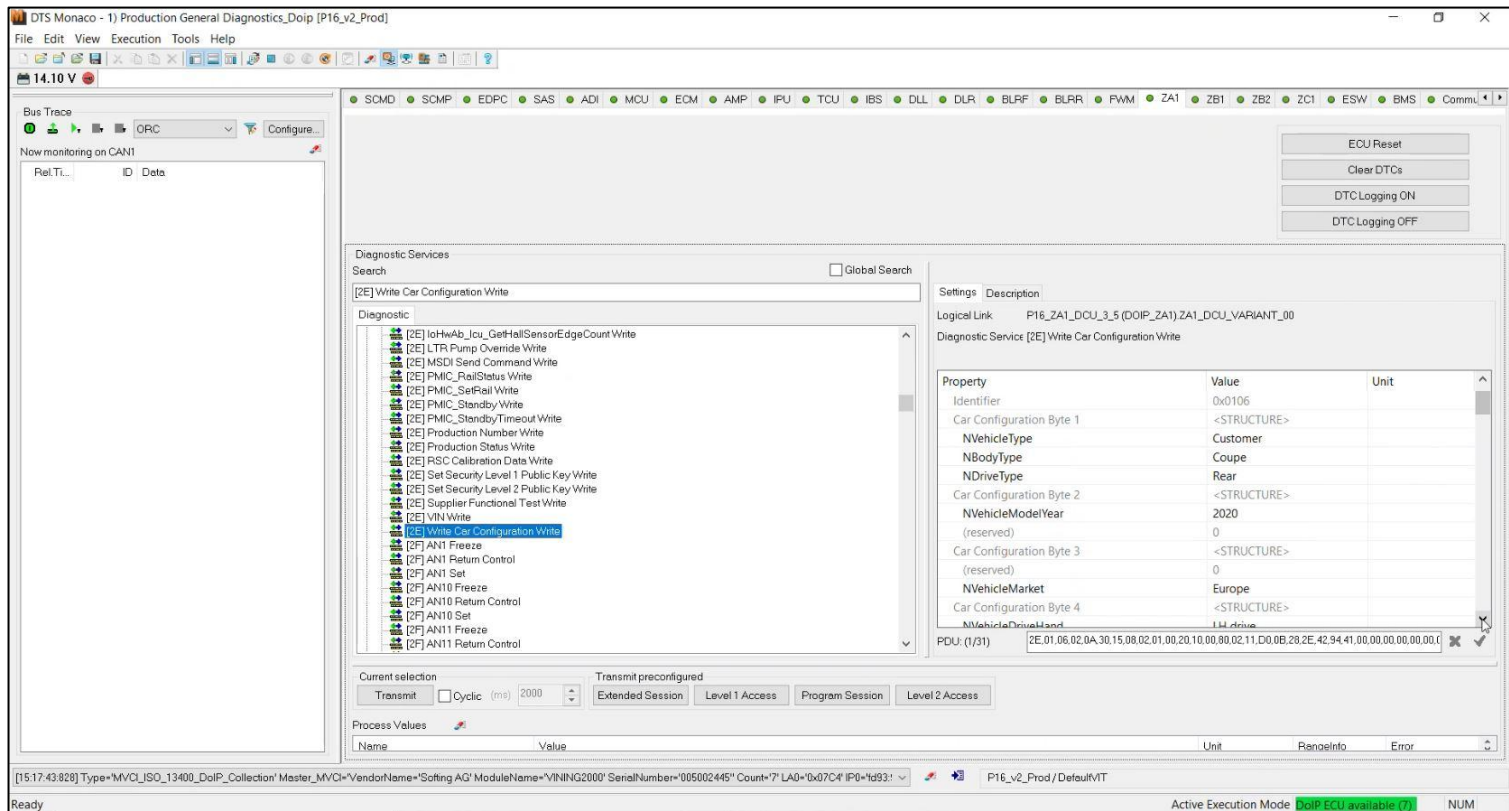
Name	Value	Unit	RangeInfo	Error
Rcode	0		Value OK	
[Tester]	[e00]			
SecurityJob	eALL_POSITIVE			
P16_ZA1_DCU_3_5 (DOIP_ZA1) Z...	[7e7]			
Response State	acknowledged			
message	Sending physical request Send_Key_Level_1		Value OK	
Rcode	0		Value OK	
[Tester]	[e00]			
SecurityJob	eALL_POSITIVE			
P16_ZA1_DCU_3_5 (DOIP_ZA1) Z...	[7e7]			
Response State	acknowledged			
message	Setting result: <<<>>> Success - Completed McLaren Security Job <<<>>>		Value OK	
Rcode	0		Value OK	

[15:17:43:826] Type="MVCI_ISO_13400_DoIP_Collection" Master_MVCI="VendorName="Softing AG" ModuleName="V'NING2000" SerialNumber="005002445" Count="7" LA0="0x07C4" IF0="Id93:"

P16_v2_Prod / DefaultMT

Ready Active Execution Mode: DoIP ECU available (7) NUM

Car config





DTS Monaco - Fault memory read

DTS Monaco - 1) Production General Diagnostics_Doip [P16_v2_Prod]

File Edit View Execution Tools Help

14.08 V

Bus Trace ORC

Now monitoring on CAN1

Rel.Tl... ID Data

Diagnostic Services

Search

Global Search

Diagnostic

- [09] ECUNAME Read InfoType Data (OBD)
- [09] Supported InfoTypes Read supported InfoTypes (OBD)
- [04] Emission-related Trouble Codes Read (Permanent Status)
- [10] Default Session Start
- [10] Default Session Start_NoResponse
- [10] Extended Session Start
- [10] Extended Session Start_NoResponse
- [10] Programming Session Start
- [10] Programming Session Start_NoResponse
- [11] Hard Reset
- [11] Soft Reset
- [14] Fault Memory Clear
- [19] Fault Memory Read By Status Mask
- [19] Fault Memory Read Extended
- [19] Fault Memory Read Number By Status Mask
- [19] Fault Memory Read Snapshot
- [19] Fault Memory Read Supported
- [22] AD Controller Inputs Read
- [22] AD Faults Read
- [22] AD Force Demands Read
- [22] AD Sensor Inputs Read
- [22] AD Valve Drives Read
- [22] ANI Read

Settings Description

Logical Link P16_ZA1_DCU_3_5 (DOIP_ZA1) ZA1_DCU_VARIANT_00

Diagnostic Service [19] Fault Memory Read By Status Mask

Property	Value	Unit
Params		
SID-RQ	0x19	
ReportDtcByStatusMask	0x02	
DTC Status Byte	<STRUCTURE>	
Test failed	<invalid>	
Test failed this monitoring cycle	<invalid>	
Pending DTC	false	
Confirmed DTC	true	
Test not completed since last clear	false	
Test failed since last clear	true	
Test not completed this monitoring cycle	<invalid>	
Warning indicator requested	<invalid>	

PDU: (1/3) 19.02.08

Current selection Transmit Cyclic (ms) 2000 Transmit preconfigured Extended Session Level 1 Access Program Session Level 2 Access

Process Values

Name	Value	Unit	RangeInfo	Error
------	-------	------	-----------	-------

[15:17:43.828] Type=MVCL_ISO_13400_DoIP_Collection Master_MVCL=VendorName=Softing AG ModuleName=VINING2000 SerialNumber=005002445 Count=7 LA0=D07C4 IP6=Id93: P16_v2_Prod / DefaultMT

Ready Active Execution Mode DoIP ECU available (7) NUM



DTS Monaco - 1) Production General Diagnostics_Doip [P16_v2_Prod]

File Edit View Execution Tools Help

14.00 V

Bus Trace
ORC
Now monitoring on CAN1

Rel.Ti... ID Data

● PEPS ● SVC ● RVC ● SCMD ● SCMP ● EDPC ● SAS ● ADI ● MCU ● ECM ● AMP ● IPU ● TCU ● IBS ● DLL ● DLR ● BLRF ● BLPR ● FWM ● ZA1 ● ZB1 ● ZB2 ● ZC1

Diagnostic Services
Search
[Z2] RSC Calibration Data Read

Diagnostic
[Z2] Rail 5V_2 Read
[Z2] Rail 5V_CAN Read
[Z2] Rail 5VA Read
[Z2] RSC Calibration Data Read
[Z2] Set Security Level 1 Public Key Read
[Z2] Set Security Level 2 Public Key Read

Settings Description
Logical Link P16_ZA1_DCU_3_5 (DOIP_ZA1) ZA1_DCU_VARIANT_00
Diagnostic Service [Z2] RSC Calibration Data Read

Property Value Unit
PDU: (1/3) 22.01.77

Current selection
Transmit ☐ Cyclic (ms) 2000
Transmit preconfigured
Extended Session Level 1 Access Program Session Level 2 Access

Process Values

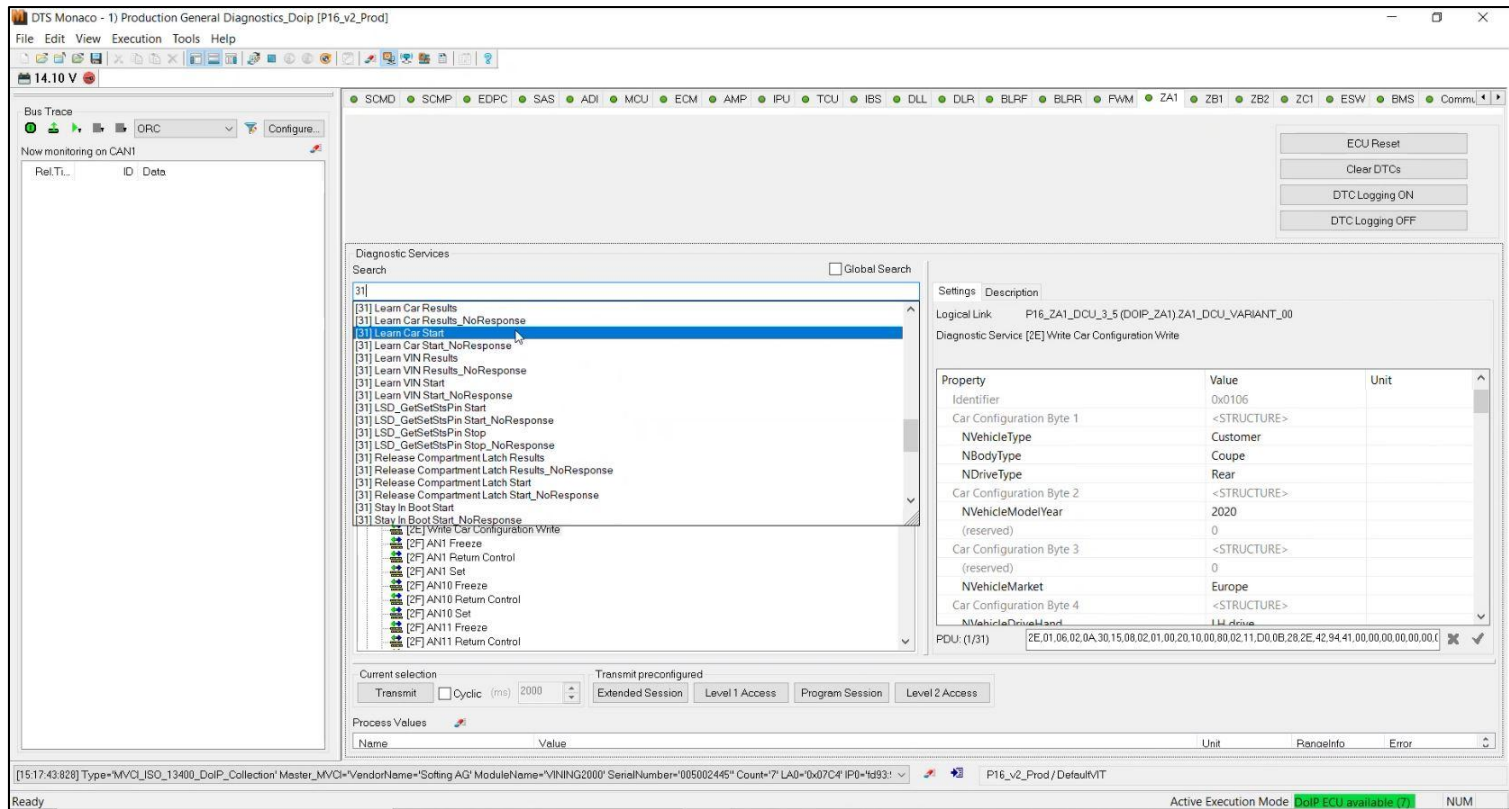
Name	Value	Unit	RangeInfo	Error
RSC Calibration Data Read	eALL_POSITIVE			
SID-RQ	0x22		Value OK	
Identifier	0x0177		Value OK	
P16_ZA1_DCU_3_5 (DOIP_ZA1) Z...	[7e7]			
Response State	acknowledged			
SID-PR	0x62		Value OK	
Identifier	0x0177		Value OK	
VxDamperFLDroopEOL	3.388	V	Value OK	
VxDamperFRDroopEOL	1.827	V	Value OK	
VxDamperRLDroopEOL	1.650	V	Value OK	
VxDamperRRDroopEOL	3.421	V	Value OK	
VxDamperFLRideEOL	2.546	V	Value OK	
VxDamperFRLideEOL	2.678	V	Value OK	

[15:36:15:188] STAT: Object with name not found; DYN: Could not find the shortname "RSC_Calibration_Data_Read" ErrorCode: ePAR_ITEM_NOT_FOUND (0x841BC020)

P16_v2_Prod / DefaultVIT

Ready Active Execution Mode: DoIP ECU available (7) NUM

Routine





DTS Monaco - Clear & Read DTC Workspace

DTS Monaco - 4) Clear DTC and DTC Read for all and Each unit on DOIP ECUs - Idiada [P16_v2_Prod]

File Edit View Execution Tools Help

14.03 V

ECUs tab menu

Read DTCs Read Data Clear DTCs ACF DML DMR EPHS HYCU SAS ESP SVC RVC SCMD SCMP ZA1 ZB1 ZB2 ZC1 PEPS ESW ORC

Bus Trace

PEPS Configure...

Now monitoring on CAN1

RelTim... ID Data

Bus Trace window

Read DTCs

Read Confirmed DTCs

Read Pending DTCs

Read Since Last Clear DTCs

Process Values

DTC Trace window

Name	Value
DoIP_ASCU_PEPS	[6c0]
Response State	acknowledged
RESPONSE_PDU	0x59.02.7F.93.37.00.2E.93.36.00.6D
SID_FR	0x59
ReportDtcByStatusMask	0x02
DTC	0x933700
Display Trouble Code	B133700
Trouble Code Text	VinMismatch
DTC Level	0
StatusOfDtc_Test_failed	false
StatusOfDtc_Test_failed_this_monitoring_cycle	true
StatusOfDtc_Pending_DTC	true
StatusOfDtc_Confirmed_DTC	true
StatusOfDtc_Test_not_completed_since_last_clear	false
StatusOfDtc_Test_failed_since_last_clear	true
StatusOfDtc_Test_not_completed_this_monitoring_cycle	false
StatusOfDtc_Warning_indicator_requested	StatusOfDtc_Test_not_completed_this_monitoring_cycle
DTC	0x933600
Display Trouble Code	B133600
Trouble Code Text	NoVinStored
DTC Level	0
StatusOfDtc_Test_failed	true
StatusOfDtc_Test_failed_this_monitoring_cycle	false
StatusOfDtc_Pending_DTC	true
StatusOfDtc_Confirmed_DTC	true
StatusOfDtc_Test_not_completed_since_last_clear	false
StatusOfDtc_Test_failed_since_last_clear	true
StatusOfDtc_Test_not_completed_this_monitoring_cycle	true
StatusOfDtc_Warning_indicator_requested	0

[15:28:08.291] STAT: System : Boundaries of intervals overlap at CompuMethod Scale Linear; DYN Error during interpretation of DTC; no matching entry found for TroubleCode 0xd80687 Error

P16_v2_Prod / DefaultVIT

Ready

Active Execution Mode **DoIP ECU available (7)** NUM



DTS Monaco - Part Number Read Workspace

DTS Monaco - 2) Read Identification Read of all and Each unit on DOIP ECUs [P16_v2_Prod]

File Edit View Execution Tools Help

14.08 V

Bus Trace

Now monitoring on CAN1

Ret Tim... ID Data

ECUs tab menu

ZB1 ZB2 ZC1 PEPS ESW ORC ARF BLRF BLRR MCU IFU IBS FWML FWMR DLL DLR ADI TCU AMP ECM BMS All Part Number Read

All Part Number Read

Part Number Read

Configure...

Process Values

Name	Value	Unit	RangeInfo	Error
Delay	eALL_POSITIVE			
[Tester]	[e00]			
Application_Software_ID_Read	eALL_POSITIVE			
SID_RQ	0x22		Value OK	
RecordDataIdentifier	0xF181		Value OK	
REQUEST_PDU	0x22.F1.81			
DOIP_ZA1_P16_ZA1_DCU_3_5	[7e7]			
Response State	acknowledged			
RESPONSE_PDU	0x62.F1.81.31.36.4D.41.31.38.35.43.50.2E.31.33.2E			
SID_PR	0x62		Value OK	
RecordDataIdentifier	0xF181		Value OK	
Application_Software_ID	16MA185CP.13		Value OK	
[Tester]	[e00]			
ID_Block_Read	eALL_POSITIVE			
SID_RQ	0x22		Value OK	
RecordDataIdentifier	0xF100		Value OK	
REQUEST_PDU	0x22.F1.00			
DOIP_ZA1_P16_ZA1_DCU_3_5	[7e7]			
Response State	acknowledged			
RESPONSE_PDU	0x62.F1.00.31.36.4D.41.37.37.35.43.58.00.00.00.31.36.4D.41.3...			
SID_PR	0x62		Value OK	
RecordDataIdentifier	0xF100		Value OK	
McLaren_Hardware_Part_Number	16MA775CX		Value OK	
McLaren_Software_Part_Number	16MA185CP.13		Value OK	
Calibration_Part_Number	16MA185CP.13		Value OK	
[Tester]	[e00]			
Delay	eALL_POSITIVE			
[Tester]	[e00]			
A123_SW_Version_Read	eALL_POSITIVE			
SID_RQ	0x??		Value OK	

Part Number Trace window

[15:31:51.543] STAT: 0x0019: DYN: Service execution not allowed on logical link in state offline ErrorCode: eRT_NOT_ALLOWED_IN_LL_STATE_OFFLINE (0x00000019)

P16_v2_Prod / DefaultVIT

Ready

Active Execution Mode BciP ECU available (7) NUM



DTS Monaco - Software Download Workspace

DTS Monaco - 3) Production Software Download Platform 4.1_DCUF8LUpdate_DOIPAIECUs [P16_v2_Prod]

File Edit View Execution Tools Help

ECUs tab menu

Bus Trace

Now monitoring on CAN1

Rel.Tim...	ID	Data
------------	----	------

Bus Trace window

ZA1

Program ZA1

Read Current Software Version

HardReset

Program ZA1 (from Boot)

Configure...

Process Values

Name	Value	Unit	RangeInfo	Error
------	-------	------	-----------	-------

Flashing Trace window

CONTACTS

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